Quick Start Guide: Accessing HDF5 Data

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Introduction

This guide explains how to open and retrieve data from any HDF5 file using Python and MATLAB, particularly the HDF5 files following the guidelines presented in the HDF5_BLS project.

In short a HDF5 file is a container for data that can be organized in a hierarchical structure. It is a great option to organize in a single file a large amount of arrays and attributes.

1 Step 1 - Simple visualization using myHDF5

Before any further step, we recommend using the myHDF5 viewer to preview the file, its components, and its hierarchy.

1. Go to myHDF5 viewer
2. Drag and drop your file
3. Navigate your file through the web application

You should see something like:

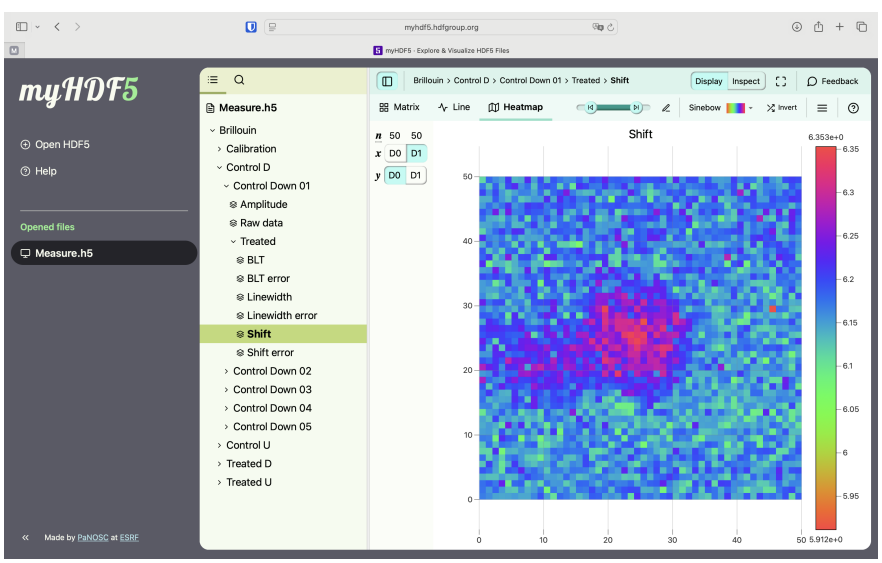

Figure 1: Example of HDF5 file structure in myHDF5 viewer

1.1 Step 2 - Locating the Dataset Path

HDF5 uses a Unix-style path system. To get the exact path for your code:

1. Select the dataset in the Tree View.
2. Navigate to the "Inspect" tab (top right)
3. Get the "Path" field (e.g., `/group_name/subgroup/dataset_name`).

For example here: the path is: "/Brillouin/Control D/Control Down 01/Treated/Shift"

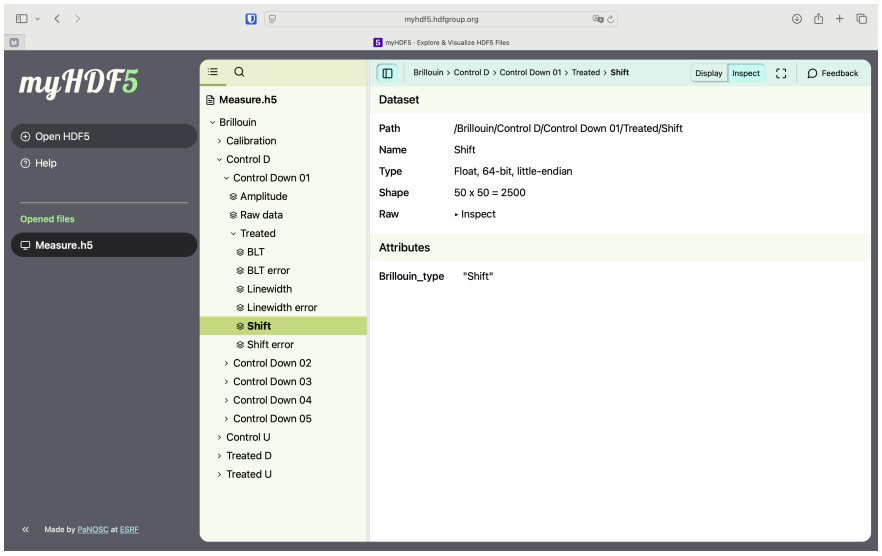

Figure 2: Example of HDF5 file structure in myHDF5 viewer

2 Step 2 - Extracting Data via Code

Once you have the path from myHDF5, use the following snippets to pull the data into your workspace.

2.1 Python (using h5py)

Python requires the `h5py` library - you can install it using 'pip' writing the following line in the terminal: `pip install h5py`.

Once h5py is installed, you can access your dataset as follows:

```python
import h5py
import numpy as np

filename = "your_file.h5"
dataset_path = "/path/to/dataset" # Obtained from myHDF5

with h5py.File(filename, "r") as f:
    # Access the dataset and convert to a NumPy array
    data = f[dataset_path][()]
```

Listing 1: Python extraction

2.2 MATLAB

MATLAB has built-in support via the `h5read` function, so no extra library have to be installed:

```matlab
filename = 'your_file.h5';
dataset_path = '/path/to/dataset'; % Obtained from myHDF5

% Read the data directly into a variable
data = h5read(filename, dataset_path);
```

Listing 2: MATLAB extraction